

AB

ABDELHAK BOURDIM

Senior Embedded Software Engineer
C/C++ • RTOS • Embedded Linux

CONTACT

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LANGUAGES

French ●●●●● *Fluent*

English ●●●●● *Professional*

Arabic ●●●●● *Fluent*

STANDARDS & METHODS

- DO-178 MISRA C IEC 62304
- ISO 15118 (V2G) ISO 14443 (Smart Card)
- ISO 15693 (RFID) OCPP V-Model
- Agile / Scrum

SKILLS

LANGUAGES
C, C++, Python, Bash, Assembly, Perl, Awk

OS / RTOS
Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel

MICROCONTROLLERS
ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto, Arduino

PROTOCOLS / BUS
CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP

STACKS
TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer

SECURITY / PAYMENT
Smart Card, RFID Mifare, DES, RC4, USB CCID

TOOLS
Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX

MANAGEMENT
Git, Synergy, Jira, Polarion, Doxygen

PROFILE

Embedded software specialist — firmware, middleware and application layers, bare metal, RTOS or Embedded Linux — for critical and industrial systems, with **over 20 years of experience** in driver development, communication stacks (CAN, BLE, USB, TCP/IP), RTOS and Embedded Linux on ARM Cortex, STM32, Microchip and NXP architectures.

Delivered certified embedded solutions for **aerospace** (Safran, Thales — DO-178), **medical** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky smart meters), **payment systems** (Cartadis) and **V2G smart charging** (E2-CAD — ISO 15118). MISRA C compliant development. Founder of **Workshop-DIY**, an educational robotics non-profit having led 18+ workshops for youth (Arduino, ESP32, AI, 3D printing).

PROFESSIONAL EXPERIENCE

E2-CAD — May 2025 – Mar. 2026 Eragny

EV Charging Station — Iotecha IP

Product takeover and test bench setup

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), setup of debug and programming tools (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, frame extraction with Saleae)
- Python extension development and real-time HTML web report generation
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Charging station debug and commissioning (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, Chargebyte, RS232, I2C, CAN, TCP/IP, Saleae, HackRF One, SSH, SCP, Wireshark, Python, Bash, HTML

Safran — Mar. 2024 – May 2025 Créteil

SSPC Project — Solid State Power Controller

Low-level driver updates

- Development and update of low-level C drivers for the SSPC system
- Implementation of logging tools for diagnostics and exchange traceability
- Writing and execution of on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

INDUSTRY SECTORS

- ✈ Aerospace — Safran, Thales, Zodiac
- 🏥 Medical — GE Healthcare, Sorin
- 🚗 Automotive — Valeo, Arkamys
- ⚡ Energy — Enedis (Linky)
- ⚡ EV Charging / V2G — E2-CAD
- 📶 IoT / M2M — Vertical M2M
- 💳 Payment — Cartadis, Photomaton, BCM

KEY STRENGTHS

- ✓ Embedded Linux & RTOS
- ✓ Embedded cybersecurity (pentest, audit)
- ✓ Embedded software development
- ✓ Communication protocols
- ✓ Testing & validation
- ✓ Diagnostics & debugging

NON-PROFIT PROJECT

Workshop-DIY

FOUNDER — CHELLES (77)

Robotics & STEM workshops for youth (ages 6-12+)
ESP32 robot kit design (WiFi, BLE, servos, sensors)
18+ workshops: Arduino, micro:bit, AI, 3D printing, laser
workshop-diy.org

EDUCATION & TECH WATCH

Electronic & Computer Systems Design Engineer
AFPA, Champs-sur-Marne

Biomedical Instrumentation (Postgraduate)
Tenon Hospital, Paris

Electronics Engineer, Communication
Boumerdès, Algeria

TECHNOLOGY WATCH

- ▶ Embedded cybersecurity & IoT pentest
- ▶ V2G / ISO 15118 / Smart Charging
- ▶ SDR & RF protocol analysis
- ▶ Embedded AI & Edge Computing

PROFESSIONAL EXPERIENCE

Thales — Nov. 2020 – Jun. 2021

Vélizy-

Villacoublay

ATO — Autonomous Train Operation

IPC communication & socket programming

- ▶ Application migration from 32-bit to 64-bit on Embedded Linux platform
- ▶ IPC communication protocol definition and implementation
- ▶ Network diagnostic tools development and integration testing
- ▶ Frame analysis with Wireshark, Tshark, Pscapy and Pyshark

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Pscapy, Python, Jira

PK Vitality — Jun. – Sep. 2020

Paris

K'Watch CGM — Continuous Glucose Monitor

BLE drivers, FreeRTOS, STM32 TouchGFX

- ▶ HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- ▶ STM32 TouchGFX touchscreen, FreeRTOS, low power mode
- ▶ In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L4S9, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

Zodiac Feb. 2017 – Mar. 2020

Aerospace

(Safran) —

Montreuil /

Vancouver,

Canada

GENOME / CORAC / SSPC A350 AC9

Low-level drivers, diagnostics, application layers

- ▶ Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- ▶ uAFDX to CAN gateway implementation and diagnostic functions
- ▶ SSPC A350 application layer development and optimization
- ▶ Validation testing and aerospace standards-compliant documentation
- ▶ Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

Sorin — Nov. 2014 – Apr. 2015

Clamart

Platinum — BLE Module for Pacemaker

Bluetooth integration feasibility study for cardiac implant

- ▶ Bluetooth Smart module validation, Serial Port Service Profile development
- ▶ Smartphone communication protocol development and Android application
- ▶ Hardware interface design and sample preparation for implantation

Env: Cortex M0, Keil, C, Android (Eclipse)

Asterios Aug. 2023 – Feb. 2024

Technologies
— Massy

T1042SOM Project — Software Tests

Software test procedure writing

- C development of validation tests for the SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing compliant with project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polaron

Thales — May 2022 – Jul. 2023
Gennevilliers

LPM + N-MMR Projects — Defense Radio

Mission preparation software & New Multi Mode Receiver

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- Development and execution of HLT (High Level Tests)
- Bug fixing and network protocol analysis (Websockets)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

Valeo — Créteil / Mar. – Apr. 2022
Egypt

Test Audit — Valeo Egypt

Unit test, integration test and tools audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation and execution time improvement proposals
- Client/system requirements review, audit report writing

Env: NI Mastre, Vector CANoe

Arkamys — Jul. 2021 – Feb. 2022
Levallois-

Perret

GStreamer Framework — Embedded Audio

Audio plugins, unit & integration tests, End-to-End tests

- GStreamer plugin and CAN driver (ELM327) development, diagnostic tools
- Unit & integration testing (Valgrind), build system Buildroot

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

General Electric 2012 – 2014

Healthcare — Buc / USA

ZYX — X-Ray Generator

CANOpen integration on RCB and CORTYX boards

- CANOpen stack integration (slave/master), CAN driver and service development
- CMX-RTX kernel porting, file management, TCL test scripts
- USA team technical support and on-site visits in the United States

Env: TI TMS320, XE167, ARM Cortex A8, VxWorks, C, C++

PREVIOUS EXPERIENCE

Enedis — May 2015 – Sep. 2016
Nanterre

Linky Smart Meters — Smart Grids

Cybersecurity and Linky concentrator audit

- Linky concentrator security diagnostics and audit
- Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- Log analysis and batch firmware update tool development
- Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Tshark, Scapy, SDR, Saleae, Python, Bash, Arduino, Raspberry Pi

Photomaton / BCM / 2010 – Vertical M2M / PGS — Paris 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

Cartadis — Oct. 2001 – Fontenay-sous-Bois Dec. 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)